

Problem Statement – Agentic Career Counseling Companion

Agent that collects user profile information such as academic background, skills, career interests, and target roles, and provides tailored career support including resume analysis, skill-gap assessment, career recommendations, roadmap generation, and interview preparation.

Description

AI Agent – CareerMentor AI is a Personalized Career Guidance Agent that collects student profile data such as educational background, current skills, career interests, and target job roles, and delivers structured, actionable career guidance. The agent analyzes uploaded resumes for ATS compatibility and improvement areas, identifies skill gaps against desired roles, recommends suitable career paths, generates phased learning roadmaps, and prepares users for interviews with structured Q&A and STAR-method guidance — all through a conversational chat interface with persistent session context.

Tools & Technologies Used:

- IBM watsonx.ai (Granite 3.3-8B Instruct model)
- Flask (Python web framework)
- Python (PyPDF2, python-docx for resume parsing)
- Flask Session (server-side state management)
- HTML/CSS/JavaScript (chat interface, marked.js for Markdown rendering)